Practice Makes Frame-Perfect: A Survey Study on the Motivations and Personalities of Speedrunners

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Universiteit van Amsterdam

Communication Science- Entertainment Communication

Abheek Dhar

12790222

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Abstract

Speedrunning is an under-researched phenomenon, which is growing every day. Speedrunning content is becoming increasingly popular to both create and watch. Despite this, there has been a lack of academic attention to it. Much of the research carried out on it focuses on a philosophical and ethical understanding of it, with almost no empirical research existing. This study aimed to understand how speedrunners interpret academic definitions of the practice, while also understanding their motivations and personality traits. The study used a combination of Self-Determination Theory (SDT) and Big 5 Item frameworks to understand the motivating factors and identifying which traits were most common amongst speedrunners

An online survey was sent out on speedrunning forums and general gaming websites. The data gathered from this set was processed using SPSS, and several statistical tests, including t-tests, logistic, and linear regressions were used to better understand the data. The results indicate that speedrunning's definition needs some amendment, and that speedrunners start practicing in order to develop relationships with people, while also feeling in control. The results further suggest that speedrunners are competitive people who generally are empathetic humans. Overall, this study acts as a basis for more research about this growing group.

Introduction

In April of 2021, Twitch Streamer and Youtuber Niftski created history in the world of gaming by becoming the first person to beat Super Mario Bros in under 4 minutes and 55 seconds. This record breaking time improved upon the previous record (which had been set by a streamer named Miniland two months prior) by a quarter of a second (Orland, 2021). The fact that Niftski had managed to break the 4:55 barrier was considered to be a monumental achievement due to the complexity and precision required to complete the game so quickly. Niftski's run was so perfect that it was just two thirds of a second away from the theoretical fastest speed run of the game (Hernandez, 2021). In order to have achieved this result, he spent hours memorizing and practicing Super Mario Bros (Nintendo, 1983). All this to break a record for a more than 2 decade old game, with little tangible reward. The concept behind speedrunning is very simple: it's trying to complete a video game as fast as possible (Lenti, 2021). The community has existed for decades and is growing online. People attempt speed runs for almost every video game imaginable. Members of the community livestream hundreds of hours of attempting to break records, and then post their record breaking runs online to speedrun.com, where they can see how they compare to other speedrunners. Over 2.5 million runs have been submitted for over 25,000 games, and that number continues to grow (About - speedrun.com, 2021).

Speedrunning is a continuously evolving phenomenon in gaming, and it has been for years. Thanks to websites like YouTube and Twitch, it has become easier than ever to stumble across someone's attempts, and learn about them. Events like "Games Done Quick", a semi-annual speedrunning charity marathon, attract hundreds of thousands of viewers (Lenti, 2021), and have raised millions of dollars. YouTubers like Summoning Salt, Smallant1, Karl Jobst, and Lowest Percent all receive millions of views on their speedrunning videos(Gray, 2021). One of the top earning Twitch streamers from last year, Ludwig Ahgren, is also a speedrun aficionado, and can be found attending conventions for them (Vox, 2018). Despite its popularity, speedrunning has received very little attention from academia.

There is a notable lack of literature related to the speedrunning subculture (Scully-Blaker, 2014). As a result, not much is known about this passionate and highly involved community. Research has attempted to understand some of the more philosophical aspects of the practice (Newman, 2019) mostly from a qualitative viewpoint. Some researchers have looked at the ethos of the subculture (Hemingsem, 2021), and there have been early attempts to understand some of the rules governing the practice itself (Scully-Blaker, 2014). Almost no papers have looked at this from a quantitative point of view, and empirical studies are scarce. There are suggestions that speedrunners tend to have depressive tendencies, be competitive, and like the community aspect of the practice, but many of these suggestions are based on limited information, or literature from other fields (Toftness, 2020). Few empirical studies have actually looked into this, and those which do exist tend to rely on anecdotal evidence, or suggestions from certain individuals. Research which actually looks at the people who partake in the community is borderline nonexistent. This thesis aims to change that.

Considering the lack of research validating the assumptions made about what drive's speedrunners, this paper looks at speedrunners' motivations and personality traits. Any notions for what propels someone to labor over a controller for hours, turning story based gameplay into a competition, is based on assumptions which have yet to be validated. Motivations in this example refers to the psychological force that enables action, or the reasons for which people carry out the action (Touré-Tillery & Fishbach, 2014). For the sake of this study, the practice of speedrunning is defined as playing a game with the intention of finishing it in as little time as possible, Practicing speedrunning, on the other hand, refers to the act of actually taking the time to learn a game inside in order to fulfill an objective. The objective in this case, is to complete a game as quickly as possible. Existing theoretical frameworks, such as Bartle’s taxonomy of gamers are ineffective for providing insight to some of the habits of that can be observed from within the community (e.g. playing the same level from a game repeatedly in order to become better) due to the fact that they define gamers in relation to the ways they interact with the story or other players (Yee, Ducheneaut, & Nelson, 2012). The dimensions outlined in frameworks like this focus tend to not have much influence in speedrunning (Scully-Blaker, 2014). This is why this study attempts to answer the question: “What characteristics and motivations do people who practice speedrunning have?”. Answering this question can help us understand this community better, which is especially important when considering how it is growing into a phenomenon which is watched and admired by millions of people online.

Theoretical Framework

The Practice of Speedrunning

Speedrunning is “an activity predicated on requiring a comprehensive mastery” as well as a practice that can help people better understand the mechanics of the games they play (Newman, 2019, p.16). As previously mentioned, literature pertaining to speedrunning is scarce, with empirical studies being even more difficult to find. There are two types of studies, those looking at the ethos and symbolic implications of the practice, as well as those looking at the habits, practices, and tactics used by speedrunners. Little to no research is available about the driving motivations behind these factors.

Literature looking at the ethos of speedrunning tends to analyse the emerging practice from a qualitative standpoint. Scully-Blaker (2014), for example, looked at the practice and places each speedrun into two categories- finesse runs and deconstructive runs. Finesse runs are defined as being video game playthroughs where the player finishes the game as fast as possible while maintaining the integrity of the story being told, while deconstructive runs abandon all sense of narrative in favour of pure speed. One example of a deconstructive speedrun is the any% speedrun, also known as fastest completion, or a category where the amount of the game completed doesn’t matter. In any% speed runs, the player is free to skip as much of the game as they would like in order to finish it as fast as possible without hacking or using external help (Newman, 2019). This category is a prime example of the deconstructive speedrun, seeing how it ignores the narrative structure of the game altogether (Scully-Blaker, 2014). An analysis of a “The Legend of Zelda: Ocarina of Time” (Nintendo, 1998) speedrun, one of the most popular games in the world, shows how players have optimised the use of glitches to skip long portions of the game (Newman, 2019) These studies are some of the earliest examples of articles looking at the practice and coming up with concrete categorisation, and even then, most of what they are based on comes from a philosophical perspective that questions the practice itself.

One perspective through which speedrunning is understood is that of high-performance gameplay (Witkowski & Manning, 2017). This framework views gaming subcultures such as e-sports and speedrunning as special forms of gaming which require extra investment on behalf of the player. Existing research has drawn a direct parallel between the activities of the subcultures, showing how the relevant practice is used as a way to learn, develop, and participate in a larger community. It is argued that context-dependent knowledge and experience are important pillars of these communities (Witowski and Manning, 2017). This is very similar to Newman's (2019) description of speedrunning being "an activity predicated on requiring a comprehensive mastery", showing that the small pool of existing research has similar observations. High skill and significant investment into practicing are recurring themes within these papers, indicating that they play a role within the practice.

One final perspective that researchers have used to examine speedrunning is by looking at the ethos, or the communal spirit of the activity. Speedrunners have very strict rules they adhere to, many of which often are extremely subjective (e.g. exploiting glitches being okay, but using cheats being unethical). These communal prerequisites have been split into 3 main parts: *skill*, *knowledge*, and *subversion* (Hemmingsen, 2021). Skill refers to the technical precision of inputs, along with the dexterity required to perform certain movements. Many speed runs require extremely precise button inputs at specific moments which requires hours of practice for many people. Mastering these movements can shave off milliseconds from each section of the game, but these perfectly timed inputs can save larger amounts of time over the duration of the run (Hemmingsem, 2021). Knowledge refers to the understanding that a speedrunner has about the game, such as the map layout and important items. It also refers to the familiarity that the player has about the code of the game, and what is possible within the way it works. The final category, subversion, refers to the way in which the code is utilised by the players, and how they get around the rules of the world which are implied (e.g. skipping a complex puzzle by jumping to avoid it) (Hemmingsem, 2021). The last category seems especially relevant for the any% speedrun due to the way people exploit glitches.

Direct parallels have been drawn between speedrunning and competition (Hemmingsem, 2021). Despite this,the same authors have pointed out a notable difference between them, arguing that speedrunning should be considered a metagame, a way of playing a game, rather than a form of gaming itself. This is due to the fact that the game should be looked at as a medium in which the speedrunners perform, rather than being a sport itself (Hemmingsem, 2021). As an extension, speedrunning, while being a form of competition, could be better compared to parkour than to competitive gaming due to the emphasis on getting around objects in the most skillful way possible. This finding implies that there should be a difference between the way that the main groups operate, seeing how the activities are different.

Overall, speedrunning is shown to be a field which requires significant investment into developing your knowledge and skill related to a game. This knowledge can include knowledge about breaking the game in a way that allows you to finish it quicker. The practice is inherently competitive, although it is not a competition in itself due to the emphasis on it being a communal practice.

Sub question 1: Are the scholarly definitions of speedrunning valid according to community members?

Motivations For Consuming Video Games

Considering the dearth of research related to speedrunners and their motivations, existing theories looking at similar fields need to be used in order to answer the research question. Much of the research on videogame play is based on Self Determination Theory (SDT) (Deci & Ryan, 2008) due to how well it explains motivations. According to SDT, there are three main types of needs that people aim to fulfill (Azadvar & Canos, 2018; Rigby and Ryan, 2016). These are *autonomy*, the freedom to make your own choices, *competence*, the need to feel capable while improving periodically, and *relatedness*, the feeling of belonging in a group (Azadvar & Canos, 2018). SDT has a history of extensive use to understand video game consumption. Research has shown that a presence of the needs for autonomy, competence, and relatedness were able to predict video game playing and future consumption (Ryan, Rigby, & Przybylski, 2006). Specifically, it was found that autonomy and competence were big influencing factors, and that fulfilling the need for competence could lead to higher self-confidence and self-esteem after playing. Recent studies have also shown relatedness to be important for people, especially with the rise of online multiplayer gaming (Sheldon & Filak, 2008, Wu, Richards, & Saw, 2014).

There seems to be a consensus between different theoretical frameworks regarding some of the driving motivations for playing video games. For example, a paper by Dalisay, Kushin, Yamamoto, Liu, and Skalski (2015) has identified 3 main factors for video game consumption similar to those outlined by SDT: social motivation, playing video games to meet people and form connections, achievement, or the satisfaction that comes with improving and overcoming barriers in games, and immersion. The first two factors are similar to the relatedness and competence needs that is identified by SDT authors. Other papers have expanded on this way of understanding motivations for playing videogames, creating longer lists of factors. Researchers have identified a long list of factors that affect the desire to play videogames (Schuurman, De Moor, De Marez & Van Looy, 2008). Many of these factors fall within the SDT identified needs.

While knowing what drives people to play video games is useful, it is not fully applicable to this study's main target group. Consequently, it is important to look at research carried out on other forms of high performance gameplay, such as e-sports (Witkowski & Manning, 2017). Research on this field reflects what has been described in the other literature looked at for this study, such as Schuurman and colleagues (2008), Dalisay and colleagues (2015) and Ryan and colleagues (2006). Existing studies have found three statistically significant motivations for e-sports participation: competition, peer pressure, and skill building for actual playing of the sport (Lee & Schoenstedt, 2011, p. 43). Following an SDT perspective, competition and skill-building would fall under the desire for competence, whereas peer pressure would fall under the desire for relatedness. Competence, autonomy, and relatedness are reflected in several different studies (Weiss, & Schiele, 2013; Martončik, 2015), showing that they can potentially be applied to speedrunning

Sub question 2: Do speedrunners try to fulfill any of the SDT needs through their practice?

Sub question 3: Which need do speedrunners try to fulfill the most through their practice?

Sub question 4: Is there a relationship between these motivations and the amount of time spent speedrunning?

Personality Traits and Gaming

Research has shown that there are many different associations which can be identified and predictions that can be made about a person based on personality traits (Azucar et al, 2018, Hayes & Joseph, 2003). Understanding what types of personality traits are attracted to speedrunning would help better understand what needs actively practicing fulfills. The most widely used of the personality frameworks is the “Big 5”: *Openness*, *Conscientiousness*, *Extraversion*, *Agreeableness*, and *Neuroticism* (Rammstedt & John, 2007). Openness refers to the individual disposition towards having new experiences. Conscientiousness refers to the presence of goal-fulfillment behavior (planning, working hard etc.). Extraversion refers to how outgoing and active a person is. Agreeableness describes how compassionate and caring a person may be for others, while neuroticism represents how sensitive/ prone to negative emotions or feelings a person can be (Chew, 2022). This framework has been used within the field of video game consumption research to show that there is a direct negative correlation between conscientiousness, extraversion, and agreeableness with internet gaming disorder (IGD), as well as a positive correlation between IGD and neuroticism (Chew, 2022), showing that these personality types can have an impact on the way that people play games. Such research has yet to be carried out on speedrunning.

Aside from these main traits, there are two more traits that would be interesting to see present in speedrunners; *competitiveness* and *grit*. Prior research has shown that the practice is competitive, while also not usually being a direct competition at the same time (Hemmingsem, 2021). Studies have also acknowledged the high amount of skill, knowledge, and training that speedrunners require. Understanding if these traits manifest themselves amongst speedrunners can help better understand them.

Competitiveness, in this case, is defined as the tendency of an individual to place themselves in an environment where they have to compete with others in order to thrive. Studies that use this definition have used this definition to identify the motives for entering a competitive environment (Bonte, Lomberdo, & Urwig, 2018). This definition of competitiveness is useful for this study seeing how it focuses on why people are drawn to competitive environments. The speedrunning community is itself a competitive environment, which means that this definition can help better understand what factors can predict involvement in speedrunning. Grit is usually defined as a combination of perseverance and passion for long term goals (Duckworth & Quinn, 2007). This measurement has been utilized as a predictor for success in many different fields (Rimfeld, Kovas, Dale, & Plomin, 2016, Pate et al, 2017), The “grittier” a person is, the more likely they tend to be successful in certain fields, as well as of commitment. Grit is usually measured by looking at two different dimensions: consistency of interest, and perseverance of effort. If these traits were to manifest themselves more strongly in speedrunners than regular gamers, it could indicate that these traits have a relationship with being a speedrunner.

Sub question 5: Is there a relationship between these motivations and the speedrunner’s demographics?

Sub question 6: Is there a relationship between a person’s personality and them being a speedrunner?

Sub question 7: Is there a relationship between the amount of time spent speedrunning, and the personality traits a person possesses?

Method

Sample

In order to answer the research question, a survey was sent out to several online speedrunning forums. The forums in question were the speedrunning subreddit, '/r/speedrun', subreddits dedicated to popular speedrunners (e.g. /r/smallant, /r/dream, /r/pointcrow, etc.), speedrunning discord servers, as well as the speedrun.com forum. These sites were chosen due to the fact that they are generally places for speedrunners to interact without being related to specific games or fads within the movement. Additionally, the survey was sent out to speedrunners, who were then asked to share it with their friends. While there are more active communities, such as Facebook groups targeting specific games, focusing on specific fan bases could lead to skewed results due to different games requiring different commitments in terms of time and skill. This was also done to ensure that the sample audience wouldn't skew too much towards playing a deconstructive run over a finesse run. Focusing on a more general audience allowed for a better image of what the overall community looked like, since controlling for each possible category would lead to there being far too many factors to control for.

Although the main aim of the study was to attract as many speedrunners as possible, the survey was also posted on '/r/gaming', one of the biggest gaming subreddits. The survey was introduced to the subreddit as a survey about gaming and speedrunning fans so that people who identify as gamers, but who may not necessarily speedrun themselves could also participate. People who had never played video games were not able to complete the survey.

While it is unclear how many people this survey reached, 667 responses were recorded. 249 of these responses were incomplete, stopping after consenting to the survey, which led to a final total of 418 collected responses, a 62% completion rate. Of the 418 final responses, 235 people identified as speedrunners, with the remaining 182 identifying themselves as normal gamers. 62.4% of respondents (*n* =261) reported being male, with 32.1% (*n* =126) being female, and with the remaining identifying as non binary (*n* =25) or not preferring to say (*n* =6).

Procedure

An online survey was conducted through Qualtrics. This tool was chosen due to the customizability of surveys, along with ease of use on different mediums such as mobile and pc browsers. The respondents were first introduced to the topic and were asked for their consent before proceeding with the questionnaire. The survey began by collecting demographic information of the gamers, namely their age, gender, and whether they identify as a gamer. This was done in order to ensure that the only people who play video games would be able to respond. People who claimed to play video games regularly were allowed to continue with the survey, while people who said they didn't play were shown the end card thanks to Qualtrics' question order logic function. This function was also used to show speedrunners and the average gamer different questions. The respondents were asked if they had, or currently do, practice speedrunning. Speedrunners would then be asked a series of questions about their speedrunning habits, while non-speedrunners would be moved straight to the section where they would be asked about their personality. This section would serve as the comparison point between the groups, and would be used to show a tangible difference between the personality traits if it were to exist.

Following the data collection, the information gathered was processed using Python. This was chosen due to the ease with which you can process large amounts of data. The dataset was cleaned in order to make the main data processing easier. Any incomplete responses, missing answers from gamers who consented to filling the questionnaire, were removed. Dummy variables, where necessary, were created, and any items that had to be reverse coded were changed. For questions which were only targeted towards speedrunners (e.g. speedrun time in on day, number of days speedrun in a week) , non-speedrunners were recorded as spending 0 hours a day, 0 times a week on the practice in order to see if there are any differences that can be measured with an increase in speedrunning time. From here, the information was exported and transferred to SPSS for further analysis. This was done thanks to the platform's ease with which certain types of analyses can be run. For the sake of this study, descriptive statistics, logistic regressions, and linear regressions, and several other statistical tests were carried out on SPSS.

Measures

Defining Factors: The first 4of these variables were based upon the characteristics of speedrunning identified in the literature, namely speedrunning as a competitive practice, speedrunning as a skillful practice, speedrunning as a practice requiring knowledge, and finally, speedrunning as a subversive act. The last factor was further split into two parts, with narrative subversion and both code subversion (glitches) being looked at through separate items. Each of these was measured by giving statements (e.g. “do you consider speedrunning a form of competition?”) and asking participants to answer using Likert scales. A strong presence of a certain answer could indicate that the community agrees on a certain characteristic.

Competence: the need to feel capable while improving periodically (Azadvar & Canos, 2018; Rigby and Ryan, 2016). This variable was measured through 6 items. These items were adapted from the UPEQ, a survey that used SDT to understand what motivates people to play competitive games (Azadvar & Canos, 2018). The items were taken from this survey and adapted to speedrunning by replacing terms such as “play”, “playing” and “the game” with terms like “speedrun”, “speedrunning”, and “the speedrun” (e.g “I practice to become better at speedrunning”, “I speedrun to challenge myself”) (*a* =.82, *M* =3.85, *SD* =.69),).

Autonomy: The freedom to make your own choices (Azadvar & Canos, 2018; Rigby and Ryan, 2016). This variable was also measured through a Likert scale with 6 items, which had similarly been adapted from the UPEQ in a similar way to the previous variable. (e.g “speedrunning makes me feel in control of the game”, “I like using glitches to find ways around barriers”) (*a* =.757, *M*=3.85, *SD* =.76).

Relatedness: The need to feel like a part of a group (Azadvar & Canos, 2018; Rigby and Ryan, 2016). Unlike the other two UPEQ factors, this was measured through only 4 items. This is because many of the items used in that survey were based on the act of playing with other people, or even to the story of the game/ characters that they may be using in the online game. These items were determined to not be relevant to the purpose of this study, seeing how speedrunning is a (mostly) single-player activity, and how the existing research shows it to be a primarily narrative-ignoring practice (Scully-Blaker, 2014) (*a* =.78, *M* =3.17, *SD* =.78).

Openness: The first of the traits outlined in the big five item list (Rammstedt & John, 2007). This variable is defined as the individual disposition towards having new experiences, or how likely a person is to try new things. This was measured using a 5 point Likert scale with two items that were taken directly from the abridged version of the big 5 questionnaire (*r* = 1.98, *M*=3.59, *SD* =.95).

Conscientiousness: The presence of goal-fulfillment behavior (planning, working hard etc.) in a person, or how committed a person is to their goals. This was similarly measured with 2 items on a 5 point Likert scale (*r* =.342, *M*=2.98, *SD*=.86).

Extraversion: How outgoing and active a person is. This is mostly defined in terms of social activity. Like the others, this item was also measured with 2 items on a 5 point Likert scale (*r*= .463. *M* =2.29, *SD* = 1.03)

Agreeableness: How caring and empathetic an individual is towards others. Like the others, this item was also measured with 2 items on a 5 point Likert scale. These items were also unaltered from their original use in the abridged scale (*r*= .2.9, *M*=3.51, *SD*=,84)

Neuroticism: How predisposed an individual is towards negative emotions/ or feelings. The fifth trait found in the big 5 item list. As with all the other big 5 personality traits, this was measured with 2 items from the abridged version of the questionnaire on a 5 point Likert scale (*r*=.467, *M* =3.28, *SD* =1.05).

Competitiveness: The inclination to place yourself in a competitive environment. This factor was looked at using the definition identified by Bonte et al's (2017) definition that was mentioned above. The items that this was measured with were based on a scale identified by the same authors, who had themselves taken the highest coded items from other scales looking at this trait (*a* =.853, *M* = 3.4, *SD* =1.01).

Grit: This factor was measured using the Grit-S, an existing scale for measuring grit (Duckworth & Quin, 2017, Rimfield et al, 2016). The items taken directly from this survey due to the precedence of it being an effective method for studying the trait. There were 8 items in total, which had been split into the two main parts- perseverance and passion. An average score of the two parts was created to determine how much grit a person had (*a*=.79, *M*= 2.69, *SD*=.659).

Weekly Speedrun Time: The amount of time spent in one speedrunning session multiplied by the number of days speedrun in a week (*M*= 7.1, *SD*=13.91).

Results

Descriptive Results

The population of speedrunners heavily leaned towards being male, with only 33.3% (*n* =41) of speedrunners identifying as female. This is different from the gamer population which had a near equal gender ratio (46%, *n* =84). A chi-square test revealed that there was a significant relationship between gender and being a speedrunner (X2(2, N=387) = 41.25, p <.001). The average length of time spent in one day of speedrunning had a mean of 2.53 hours (*SD* =2.45). The data shows that many of the participants were new to speedrunning, with 37% of respondents (*n* =108) having started speedrunning within the last two years, and only 4.2% (*n* = 18) having started more than 5 years ago (6+ years). The most popular games to speedrun, amongst the sample, were Minecraft (Mojang, 2011), (*n* =89), various titles from the Mario franchise (Nintendo, 1981) (*n* =43), and various titles from the Zelda franchise (Nintendo, 1986) (*n* =33). As a group, speedrunners spent 1.98 days speedrunning in a week (*SD* =2.07). The mean weekly time spent speedrunning was 7.1 hours (*SD* = 13.91), indicating that speedrunners with different levels of dedication were reached. A t-test was conducted to see if there was a relationship between the weekly amount of time spent speedrunning among male (*M*= 7.1, *SD*= 12.24) and female(*M*= 7.71, *SD*= 20.18). No significant relationship was identified between the speedrunning habits males and females who identified as speedrunners(*t* (184) = 1.21, p = .272).

Community Interpretations of Speedrunning

While there is a lack of research about speedrunning, what little research exists has focused on defining or understanding the act (Scully-Blaker, 2014, Hemingsem, 2021, Newman, 2019). Academics seem to agree on a couple key parts of what defines the community, but there is no evidence looking at whether the people who practice agreed with those definitions. Speedrunners agreed that the practice is inherently competitive, with the mean being 4.55 (*SD* =.65). This was the defining characteristic with the highest mean and lowest standard deviation, indicating a high amount of agreement. Speedrunners similarly seemed to agree that knowledge and skill were parts of the practice, having means of 4.32 (*SD* =.93) and 4.33 (*SD* =.97) respectfully, with the higher standard deviations implying less agreement. The items about subversion also had similar responses. Code subversion's results were similar to the other traits, with a mean of 4.24 (*SD* = .96). Narrative subversion's results were less unanimous, with a mean of 3.98 (*SD* =1.15). The score, along with the higher variance in the results show that a surprisingly high number of people believe that game narratives are important for speedruns. A factor analysis revealed two factors with eigenvalues>1. Competition, knowledge, and skill belonged to the first factor, with narrative and code subversion belonging to the second. This indicates that subversion isn't a defining characteristic of speedrunning, but rather a means to the end.

Table 1

Correlations of speedrunning's defining characteristics

	Mean, SD	*Competition*	*Knowledge*	*Skill*	*Code*	*Narrative*
Competition	4.54, .64	*1*				

Knowledge	4.33, .97	*.16**	*1*			
Skill	4.28, .92	*.20***	*.53****	*1*		
Code	4.24, 0.96	*.09*	-*.19***	-*.11*	*1*	
Narrative	2.26, 1.16	-*.02*	*.08*	*.01*	-*.33****	*1*

*Note: * p < .05, ** p < .01, *** p < .001*

The results of the factor analysis matched the findings of a series of bivariate correlations. The correlations showed significant differences between competition, knowledge and skills, showing that speedrunners agree on these three factors being defining characteristics of speedrunning. Code subversion (the use of glitches), was negatively correlated with the importance of narratives in the speedrun (*p* <.001*)*, implying that people who favor the use of glitches, and those who like following the story in a speedrun are two different groups. Code subversion's negative relationship with knowledge is interesting, seeing how it implies that people who believe knowledge is an important part of speedrunning do not agree with the use of glitches, and vice versa, contradicting the results. This, along with the fact that it doesn't correlate with the remaining variables, nor fall into the same category according to the factor analysis, indicates the community doesn't agree on it being a defining characteristic.

Speedrunning and Motivations

Three multiple regressions were conducted in order to understand if there was a relationship between the strengths of these motivations and their speedrunning habits and history.

Table 2

Correlations between Weekly Time on Speedrunning and Personality Traits

	Weekly Speedrun Time	Years of Speedrunning	Age
Competence	$r = .27, p < .001$	$r = .24, p < .001$	$r = .01, p = .048$
Autonomy	$r = .32, p < .001$	$r = .25, p < .001$	$r = .07, p =.14$
Relatedness	$r = .27, p < .001$	$r = .34, p < .001$	$r = .08, p =.11$

Two regressions were carried out for this; one against their weekly time spent speedrunning and one against the number of years they have been speedrunning. Each of the regressions showed significant positive correlations between the motivations and how many hours someone speedruns in a week, $R^2 = .125$, $F(3, 414) = 19.7$, $p < .001$, as well as the number of years someone has been speedrunning, $R^2 = .112$, $F(3, 414) =17.4$, $p <.001$. While the former model was significant as a whole, the only motivation which had a significant impact was relatedness ($B =3.47$, $p<0.01$). This indicates that the relatedness is the most impactful of these motivations, even though the rest of these motivations are strongly spread out amongst the community. For the model comparing motivations to the number of years spent speedrunning, Autonomy ($B =.69$, $p<0.01$) and relatedness ($B =.34$, $p=.046$) were shown to be significant motivators. This means that the communal aspect of speedrunning has been a factor drawing people to the phenomenon for years.

Aside from these, another multiple regression was conducted to understand the relationship between the motivations and the speedrunner's age. The regression resulted in an insignificant result ($R^2 = .02$, $F(3, 414) = 1.42$, $p = .24$), implying that there is absolutely no relationship between the age of a speedrunner and their motivations. A regression was also held for understanding the relationship between these motivations and gender (specifically being male or female). For this example, dummy variables were created where all men were marked as 1, and women as 0, all others were treated as missing. This regression had significant results ($R^2 = .065$, $F(3, 383) = 8.87$ $p < .001$), implying that gender and motivations are positively, thought also marginally, related. Despite the model's significance, none of the motivations had significant relationships with gender.

Personality Traits- Gamers vs Speedrunners

Bivariate correlations were carried amongst all the personality traits found in order to understand if there were any traits that may not fit into the data. The correlations showed several significant negative correlations between neuroticism and grit, competitiveness, conscientiousness, conscientiousness, and agreeableness. These show that respondents who show signs of being prone to negative thoughts are less likely to place themselves in competitive environments, display goal oriented, or show concern for other people. The other significant relationships identified were between conscientiousness, and both grit and openness. The strong positive relationship with grit shows that people who display more goal oriented behavior have more passion and perseverance. The positive relationship with openness shows that these individuals are also open to new experiences.

A logistic regression was carried out comparing these personality traits to whether people identified as a speedrunner or not. Gender and age were also used as variables in order to estimate if they had an impact or not. The model generated was significant nt, $X2$ (9, N = 418) = 58.23, $p < .001$. The model explained 21.2% (Nagelkerke R2) of people being speedrunners and correctly classified 67.8% of cases. The only statistically significant and impactful variable from the model was competitiveness (p<.001). An increase in this trait was shown to lead to an increase in the likelihood that a person is a speedrunner, indicating that there is a relationship between these two variables.

A multiple linear regression was conducted looking at the relationship between the personality traits and the length of time spent speedrunning in a week. The regression was significant, (R^2 = .073, F(6, 332) = 4.33, p =< .001). The only traits to have a significance with the amount of time spent speedrunning were competitiveness (B = 1.44, p<.015) and agreeableness (B = 1.79 p=0.09). Both of these variables had positive relationships with the amount of time spent running in a week, implying that being more competitive and agreeable as a person are related with an increased amount of time spent speedrunning.

T- tests were conducted between being a speedrunner (using a binary scale with speedrunner = 1) and competitiveness (M = 3.4, SD =1.01)) and agreeableness (M=3.51, SD=,84) due to their significant relationships with weekly speedrun time. The test for competitiveness revealed a significant relationship with being a speedrunner (M = 3.4, t (337) = 4.23, p = .006), as did the test for agreeableness (M=3.51, t (416) = 2.16, p = .031). Competitiveness' larger t value indicates that there is a larger difference in competitiveness between speedrunners and non speedrunners than there is for agreeableness. Both results indicate that there is a difference in the strength of these personality traits between speedrunners and non speedrunners. T-tests were also conducted for the other personality traits, however, none of the results were significant.

Discussion

This paper acts as a foundation for the future of speedrunner research. This paper validates academic's existing interpretation of the practice, while also providing insight into the people who participate in practicing it. Speedrunners agreed that the practice was inherently competitive, and that knowledge and skill were also core components. They also agreed that subverting the code of a game was an important part of speedrunning, but seemed to have mixed interpretations of narrative subversion. The lack of a correlation between the subversion measurements with the rest of the variables could indicate that these factors come secondary to the other three defining characteristics. These results contradict earlier papers which claim this subversion to be core characteristics of speedrunning (Newman, 2019, Hemmingsem, 2021). Competition, knowledge, and skill being central to speedrunning makes sense, seeing how speedrunners improve by practicing techniques and methods identified by others in order to finish a run faster. The conflicting correlation between code and narrative subversion shows that the community does not hold either characteristic to be a defining characteristic of the practice, which can be seen considering how any% and 100% are the two most common speedrunning categories. An updated definition could focus on those three elements being at the core of the activity, with subversion being a result of competitive nature rather than an important part of the practice. Subversion can be seen as a means to the goal (finishing a game as quickly as possible), rather than a defining factor of the process (Scully-Blaker, 2014)

Speedrunners' motivations were also found to have a relationship with how much time speedrunners spend practicing in a week, as well as the number of years that a speedrunner has been practicing for. Relatedness was found to have significant relationships with both factors, while autonomy was related to the number of years spent speedrunning alone. These results show that the feeling of belonging to a community, can be a predictor of why someone starts speedrunning. Relatedness being the most significant motivator makes sense, seeing how speedrunning is a community oriented activity. This matches what recent studies have been able to show about online gaming, which tends to also be a community oriented activity (Wu, et al, 2014). Autonomy on the other hand, shows that the speedrunners like to be in control of the way that they play. The fact that competence was not found to significantly impact a person's weekly speedrun time is surprising, seeing how speedrunning is considered to be a skill oriented task according to the respondents. This indicates that feeling good at the game is less important than being in control of it.

This paper also attempted to identify the relationship between several personality traits and the amount of time spent speedrunning, as well as the differences between speedrunners and non-speedrunners. The results showed that there seemed to be only two personality traits which had an impact on the weekly speedrun time; competitiveness, and agreeableness. T-tests revealed that both of these personality traits were also more present in speedrunners than non-speedrunners The former trait is in line with expectations, considering how the participants agreed that speedrunning is a competitive task. Agreeableness, on the other hand, was surprising to see, considering the fact that speedrunning is (in most circumstances) a solo act. Being caring towards other human beings isn't usually something that would normally be considered necessary for solo gamers, but this trait could explain why this community is regarded as being tightly knit. The lack of any strong relationship with neuroticism shows that speedrunners aren't necessarily people who are prone to depression or anxiety, which contradicts some of the notions held about the community (Toftness, 2020). The lack of a significant correlation with openness and extraversion also shows that speedrunners also don't need to only be strict introverts who appreciate their comfort zone, or extroverts who like new experiences. These traits also had no correlation with the speedrunners' practicing schedule, history, or even their age or gender. There is still a chance that presence and strength of certain traits could predict success in speedrunning, seeing how the big 5 item list has been used to prove similar connections in previous studies (Rimfeld et al, 2016)

In spite of all these findings, there are certain limitations to this study. The most prominent limitation of the research conducted is that the data it relies on is self reported. While the participants would ideally have no reason to lie during this survey, there is no guarantee that they didn't. Research held in experimental conditions could show other results due to the fact that the information recorded would be easier for the researchers to control. Another limitation of this paper is the reliance on convenience sampling. The fact that this community is so hard to reach makes this method necessary for a survey. Simultaneously, this made it so that controlling the population, and accounting for things such as the types of games they play, country based demographics, and even the speedruns categories they practice were difficult to keep track of. Certain games have speedrun categories which are more popular to run than others which can create population imbalances that are hard to control or account for. Sometimes, there can also be differences in the version of the game being played (e.g. a specific patch of the game) which adds to this confusion. These minute details are controversial within the games community, and are subject to change often, which make it difficult to control .The final major limitation of this study is the imbalance in the population size of speedrunners vs regular gamers. Despite the difference in sizes between the two communities, speedrunners were much more enthusiastic about participating in the studies, leading to them having a much higher response rate. Future studies could attempt to replicate this study with sample populations that are closer to each other to see if the results stay similar.

While we do know now that speedrunners tend to be more competitive and agreeable than the avid gamer, we still are not sure about what motivates them truly. Future research should use different motivational frameworks to see if they do a better job of explaining speedrunners. Other examples of empirical research could also be employed, including quantitative content analyses of speedrunning forums, or even analyses of popular speedrunners. As mentioned earlier, it would also be interesting to see if there is a difference between speedrunners who have any significant records, and the average speedrunner, to see if there are any traits that predict success. There is also a lot of scope for qualitative research within this area. Content analyses of speedrunner interviews could be a good potential avenue to better understand what their core motivators are from a perspective that isn't limited by any theoretical frameworks. Ultimately, there is a large amount of potential for research in this domain due to it being mostly unexplored.

This study aimed to understand more about speedrunners, such as what motivates them, as well as what personality traits they have. This paper's main contributions to the existing pool of literature are the identification of which motivations are strongest related to being a speedrunner, as well as understanding which personality traits are most associated with being a speedrunner. This study also suggests that the way previous papers have analysed speedrunning needs to be reconsidered. In general, it can be understood that speedrunning is a competitive practice which requires lots of knowledge and skill. People who participate in this practice tend to be looking to fulfill their intrinsic needs for feeling like part of a community, as well as wanting to feel in control.These people also tend to be very competitive, and have agreeable personalities. The findings of this study shed light on the members of a growing and under researched community acting as a base to build more research on. With the rate that speedrunning is growing, knowing why people are drawn to this form of superplay over others could help identify target audiences for games that are made to be speedrun. It can also be used as a precursor for understanding what effects speedrunning may have on practitioners. Ultimately, speedrunning is still an under-researched field, and this study is by no means conclusive. With the millions of people practicing speedruns, and even more watching speedrunning content, the field deserves more academic attention. This paper acts as a step in that direction.

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